

Elapids

PHYLUM	Chordata
CLASS	Reptilia
ORDER	Squamata
SUBORDER	Serpentes
SUPERFAMILY	Caenophidia
FAMILY	Elapidae
SPECIES	361

Although elapids account for less than one in 10 of the world’s snakes, all of them are venomous—many dangerously so. They include cobras, coral snakes, mambas, kraits, and sea snakes, and are found mainly in the tropics and in the Southern Hemisphere. There are ground dwelling and semiburrowing species of elapids, as well as some that live and hunt in trees. Some are fast-moving, diurnal hunters, while others are more secretive, hunting under the cover of darkness. Sea snakes and sea kraits are wholly marine. Elapids are related to colubrids (see pp.393–8) and look superficially similar, but they differ from most of them in having fangs located towards the front of the upper jaw. This allows them to inject venom in a sudden and deadly strike.

Anatomy

Elapids typically have a slender, roughly cylindrical body, with smooth, shiny scales. Sea snakes—sometimes classified in a family of their own—are noticeably different, with a flattened tail for swimming. Most terrestrial elapids have camouflage colors and markings, but coral snakes have bold

markings that warn would-be predators to leave them alone. Some cobras intimidate their enemies by raising the front part of their bodies off the ground, and expanding their ribs to form a hood.

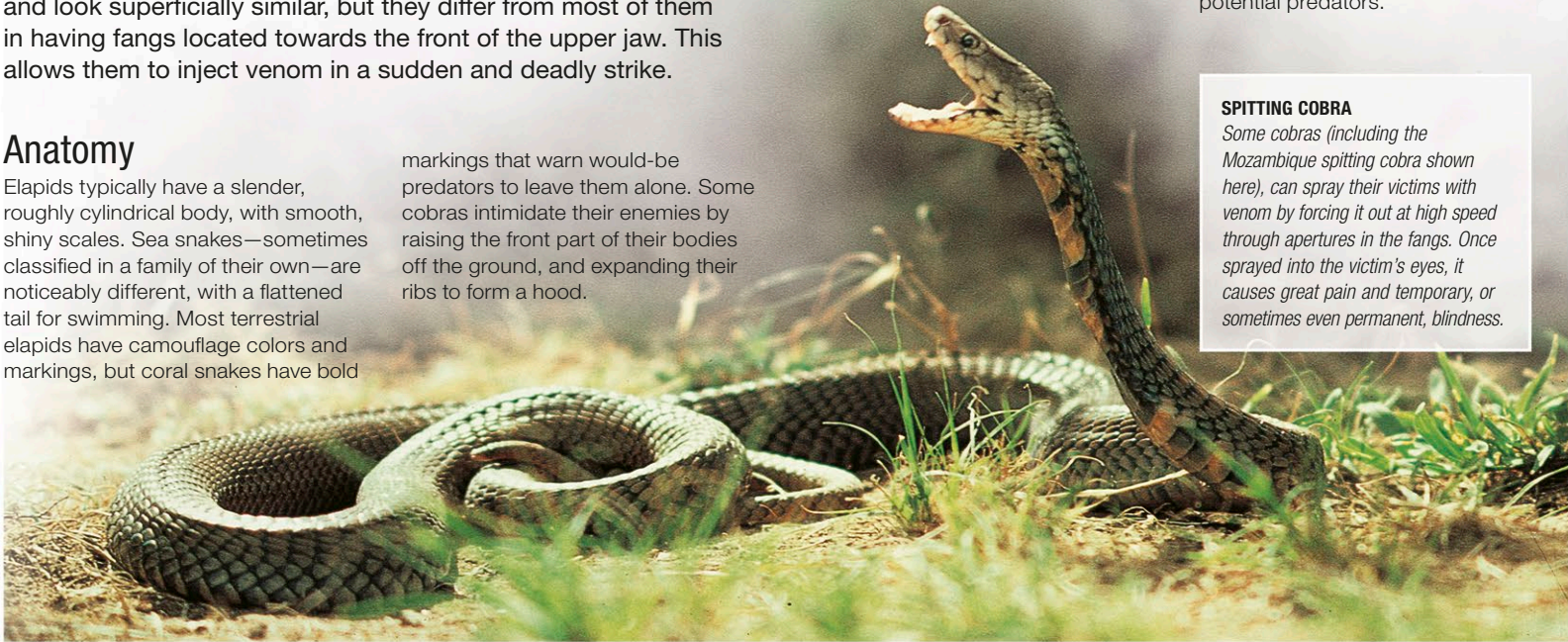
Venom

The venom of elapids varies in its potency and the organs and body systems that it affects. Most elapids, including cobras, release venom that acts on the nervous system, paralyzing muscles used in respiration. In some elapids, the venom runs down grooves in the fangs, while in others, the fangs contain an internal

venom canal. The elapids include some of the world’s most dangerous snakes, such as the taipan. Others, however, are inoffensive and too small to endanger humans.

Reproduction

Most elapids lay eggs but a few are bearers of live young. Sea snakes, for instance, give birth in the water, whereas sea kraits come ashore to lay eggs. A number of Asiatic cobras build nests out of dead leaves and other forest debris. Females guard the eggs fiercely, protecting them from potential predators.



SPITTING COBRA
Some cobras (including the Mozambique spitting cobra shown here), can spray their victims with venom by forcing it out at high speed through apertures in the fangs. Once sprayed into the victim’s eyes, it causes great pain and temporary, or sometimes even permanent, blindness.

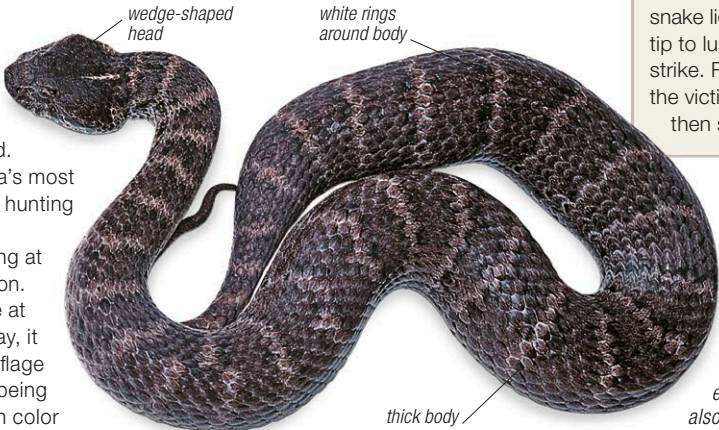
Acanthophis praelongus

Northern death adder

Length 12–39 in (30–100 cm)
Breeding Viviparous
Habit Terrestrial
Status Not evaluated

Location New Guinea, N. Australia

Despite its name, this snake is not a true adder (see pp.402–3), but an unusually stout-bodied elapid. It is one of Australia’s most venomous snakes, hunting by a sit-and-wait strategy, and striking at the least provocation. It is normally active at night; during the day, it relies on its camouflage markings to avoid being seen. Adults vary in color



from light gray to gray-brown or almost black, with paler rings or bands. Northern death adders feed mainly on small mammals, lizards, and birds, and they use their thin, wormlike tails to lure animals within striking range. They give birth to live young, producing litters of up to 8 each time they breed. A very similar species, the southern death adder (*Acanthophis antarcticus*), is widespread in eastern and southern Australia, and is sometimes found close to urban areas. Like its northern counterpart, it has a bite that is potentially fatal to humans.

LIVING LURE

The tip of the northern death adder’s tail is very slender, and is considerably paler than the rest of the body. When hunting, the snake lies still, and wriggles the tail tip to lure prey close enough to strike. Rapid-acting venom kills the victim, which the snake can then swallow at leisure.

ADDER LOOK-ALIKE
A thick body, wedge-shaped head, and narrow neck are all features normally seen in true adders and vipers, rather than in elapids. The northern death adder also has raised “horns” over its eyes.

Austrelaps superbus

Australian copperhead

Length 4¼–5½ ft (1.3–1.7 m)
Breeding Viviparous
Habit Terrestrial
Status Not evaluated

Location S.E. Australia, N. Tasmania

This inhabitant of marshes and swamps is active both during the day and at night, and feeds mainly on frogs. It is gray or reddish brown to black above, and cream or yellow on the flanks. Although it hibernates in winter, it can tolerate low temperatures and is active longer than other reptiles.

